

REUBEN SAMSON RAJ

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[Website](#) · [LinkedIn](#) · [Google Scholar](#)

EDUCATION

- Ph.D. in Computer Science**, University of Arkansas, Fayetteville, AR Aug 2021 – Dec 2026 (Expected)
Dissertation: *Frameworks on Network Programmability and Abstraction for Securing and Strengthening Smart-Grids*
Advisor: Dr. Dong (Kevin) Jin
- M.S. in Computer Engineering**, University of Arkansas, Fayetteville, AR Aug 2021 – Dec 2024
- B.E. in ECE**, College of Engineering, Guindy, Anna University, Chennai, India Jun 2006 – May 2010

RESEARCH INTERESTS

Secure and Resilient Cyber-Physical Systems, Cloud-native Systems, AI-driven Security

PUBLICATIONS - CONFERENCE PAPERS

1. Haden Fowler, **Reuben Samson Raj**, Dong Jin. *Dynamic Data-Driven Cyber Defense for Industrial Control Systems using P4-Based Cross-Layer Telemetry*. DDDAS 2026 (Accepted; to appear)
2. Yuexin Chen, **Reuben Samson Raj**, Xiaoyu Guo, Dong Jin. *Enhancing Security of Grid Control Systems through In-network Collaboration of Programmable Data Planes*. IEEE SmartGridComm 2026 (Accepted; to appear) [🔗](#)
3. Zhiyao He*, **Reuben Samson Raj***, Yanfeng Qu, Dong Jin. *Programmable Data Plane Approaches for ML-Based Anomalous Event Classification in Grid Sensor Networks*. IEEE ICC 2026 (* = Equal Contributions) [🔗](#)
4. Sarvesh Bidkar, Rakesh Abbireddy, Maryam Amiri, **Reuben Samson Raj**, Anil Rana, Jeff McLaird, Paul Rea, Meryem Simsek, Jesse E. Simsarian. *Demonstration of Multi-Provider Network and Cloud Service Provisioning with Blockchain Smart Contracts*. ECOC 2025 [🔗](#)
5. **Reuben Samson Raj**, Dong Jin. *Leveraging Data Plane Programmability Towards a Policy-driven In-Network Security Framework for Industrial Control Systems*. ACM e-Energy 2025 [🔗](#)
6. **Reuben Samson Raj**, Dong Jin. *Dynamic Data Driven Security Framework for Industrial Control Networks using Programmable Switches*. DDDAS 2024 [🔗](#)
7. Zhiyao He, Yanfeng Qu, Gong Chen, **Reuben Samson Raj**, Hui Lin, Dong Jin. *Towards Secure and Resilient Synchronphasor Networks Using P4 Programmable Switches*. IEEE GreenTech 2024 [🔗](#)
8. **Reuben Samson Raj**, Dong Jin. *A Framework to Evaluate PMU Networks for Resiliency Under Network Failure Conditions*. IEEE SmartGridComm 2022 [🔗](#)

PUBLICATIONS - BOOK CHAPTER

1. **Reuben Samson Raj**, Dong Jin. *A Dynamic Data-Driven Framework for Enhancing Energy System Security and Resilience through Programmable Networks*. Book Chapter - DDDAS Handbook Vol. IV, Springer, 2027 (Accepted; under Production) [🔗](#)

WORK EXPERIENCE

Sr. Graduate Research Assistant

University of Arkansas

Aug 2021 – Present

Fayetteville, AR

- Research on Network Security for Smart Grid and Industrial Control Networks using P4-programmable switches (Intel Tofino, bmv2) and graph-based modeling; produced multiple first-author publications
- Go-to person for P4-related issues and advising lab peers on technical solutions
- Mentor undergraduate students on thesis projects and coordinate high-school programming contests as liaison between platform vendors and organizers
- Recognized by advisor for research independence, leadership, and for stepping in to teach and moderate class discussions

Software Engineer

Cisco Systems

Jun 2020 – Jul 2021

Bangalore, India

- Performed integration and end-to-end testing for Cisco's Cloud-Native 5G Core network functions – SMF, UPF, SGW
- Deployed and managed Kubernetes-based software deployments in production environments; uncovered several critical bugs on the RADIUS interface and led the automation strategy for it

Member of Technical Staff

Parallel Wireless

Oct 2018 – May 2020

Bangalore, India

- Performed QA validation for Parallel Wireless' gateway products – TWAG (WiFi gateway) and SAEGW – including VoWiFi call scenarios over GRE/IPsec tunnels and IKEv2 tunnel establishment
- Developed Python automation for SAEGW Gx-interface tests, reducing manual test execution time by 60%

R&D Engineer

Nokia Networks

Apr 2015 – Oct 2018

Bangalore, India

- Performed software validation for Nokia's Home Subscriber Server (HSS) and 5G UDM Network Function; hands-on experience bringing up Virtual Network Functions (VNF) on OpenStack and Nokia CBAM (Cloud Band Application Manager)
- Led test strategy for the HSS 5G VNF, improving defect detection before customer deployments and reducing field issues

Software Engineer

Cisco Systems

Apr 2012 – Apr 2015

Bangalore, India

- Performed software verification on Cisco StarOS ASR 5000, ASR 5500 SGSN gateway nodes
- Developed and maintained automated regression test suites for SGSN features using TCL scripting, enhancing Cisco's test automation framework for improved efficiency

Associate Software Engineer

Accenture

Apr 2011 – Mar 2012

Chennai, India

- Performed software testing for a variety of credit card delinquency applications for Bank of America

INTERNSHIP

Network Architecture Research Intern

Nokia Bell Labs

Jun 2023 – Aug 2023

Murray Hill, NJ

- Deployed Kubernetes clusters on Nokia Airframe servers to simulate multi-cloud environments
- Optimized a YOLOv7-based object detection pipeline by converting it into a FastAPI inference service, reducing RTT latency from 500 ms to 30 ms
- Integrated NVIDIA GPU plugins with Kubernetes to improve GPU utilization for inference pods
- Scaled the video inference system, enabling real-time webcam feed processing with OpenCV

TEACHING EXPERIENCE

CSCE 47503 Computer Networks — Substitute Instructor

Fall 2025

CSCE 47503 Computer Networks — Teaching Assistant, Substitute Instructor

Fall 2026

CSCE 57903 Advanced Network Security — Substitute Moderator

Spring 2026

ACADEMIC AWARDS & HONORS

- *Blanche Bledsoe Rosecrans & Clarence J. Rosecrans, Sr. Memorial Scholarship* 2026
- [Second place - Best Paper Award](#), DDDAS Conference 2024
- [Finalist, Intern of the Year](#), University of Arkansas College of Engineering 2024
- *Student Travel Grant, ACM e-Energy* 2025
- *Larry & Gwen Stephens Graduate Scholarship* 2024
- *Reginald R. Barney & Jameson A. Baxter Graduate Scholarship* 2023
- *Department of Electrical Engineering and Computer Science Scholarship* 2022
- *21st Century Research Leadership Chair Scholarship* 2023

INDUSTRY AWARDS & HONORS

- *You Amaze and You Inspire Awards*, Cisco 2020
- *Best Feature Team Award*, Nokia 2016
- *CAP Award*, Cisco 2013

ACADEMIC SERVICE

- Journal Reviewer - *ACM Transactions on Modeling and Computer Simulation (TOMACS)* 2025, 2026
- Conference Reviewer - *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)* 2026

MENTORING & STUDENT SUPERVISION

- **Joseph Wilkin** | *Honors Thesis*
Role: Guided research formulation, methodology design, and tracked milestones via biweekly reviews.
Outcome: Thesis [published on ScholarWorks](#).
Current Position: Software Engineer, Walmart.

- **Evan Bonar** | *Honors Thesis*
Role: Advised on research formulation; provided hands-on programmable-hardware debugging support.
Outcome: Thesis [published on ScholarWorks](#).
Current Position: B.S. Alumnus (Summa Cum Laude, Fall 2024).
- **Haden Fowler** | *Honors Thesis & Conference Publication*
Role: Supervised thesis research on P4-based closed-loop defense for OT/ICS networks and co-developed the core framework and manuscript for [DDDAS 2026](#).
Outcome: Thesis [published on ScholarWorks](#); paper accepted.
Current Position: Ph.D. Student, Computer Science, University of Arkansas.
- **Yuexin Chen** | *Research & Conference Publication*
Role: Provided strategic research guidance on P4 networking tools, narrative structuring, and manuscript editing.
Outcome: Paper achieved high acceptance scores (5/6 across all reviewers) at [IEEE SmartGridComm 2026](#).

VOLUNTEERING & SERVICE

- **Lead Organizer, [Statewide High School Programming Contest \(HSPC\)](#):** Spearheaded event logistics and served as the primary vendor liaison for three consecutive years (2024, 2025 and 2026); ensured smooth technical execution of this regional coding competition
- **Exceptional Service Recognition:** Publicly commended by my advisor to the Dean of Engineering as the “*hero behind the programming contest*” for consistently exceeding expectations and ensuring smooth event operations.

SKILLS

Languages	Python, P4 (Tofino, bmv2), C, Bash
Networking	TCP/IP, SDN, 5G Core, Mininet
Cloud & Orchestration	Kubernetes, Docker, OpenStack
Security & ICS/OT	Modbus, DNP3, Network Security, IDA Pro, PowerWorld Dynamic Studio
Tools & Frameworks	Linux, Git, FastAPI, OpenCV, NVIDIA GPU Stack